

Spheroidal Harmonics

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1 Prolate Spheroidal Coordinates

We will use notation as consistent as possible with the Matlab code.

$$\begin{aligned}x &= a\sqrt{u^2 - 1}\sqrt{1 - v^2} \cos \phi \\y &= a\sqrt{u^2 - 1}\sqrt{1 - v^2} \sin \phi \\z &= auv\end{aligned}$$

$$\begin{aligned}0 &\leq \phi \leq 2\pi \\-1 &\leq v \leq 1 \\u &> 1\end{aligned}$$

The variable u determines the eccentricity of the spheroid; specifically, the eccentricity is $\epsilon = \frac{1}{u}$. Thus, when u is close to 1, the spheroid is more eccentric, but as $u \rightarrow \infty$, the spheroid becomes more spherical.

Cross-sections in the z direction are circles parameterized by angle ϕ . Cross-sections in the x or y direction are ellipses with semi-major axis $c = au$ and semi-minor axis $b = a\sqrt{u^2 - 1}$.

1.1 Differential Operators

The tangent vectors on a spheroid of fixed u and a are

$$\begin{aligned}\vec{v} &:= \frac{\partial \vec{x}}{\partial v} = \left[-a\sqrt{\frac{u^2 - 1}{1 - v^2}}v \cos \phi, \quad -a\sqrt{\frac{u^2 - 1}{1 - v^2}}v \sin \phi, \quad au \right]^T \\ \vec{\phi} &:= \frac{\partial \vec{x}}{\partial \phi} = \left[-a\sqrt{u^2 - 1}\sqrt{1 - v^2} \sin \phi, \quad a\sqrt{u^2 - 1}\sqrt{1 - v^2} \cos \phi, \quad 0 \right]^T\end{aligned}$$

Hence, the normal vector to the surface of the spheroid, which we define as $\vec{u}$, is

$$\begin{aligned}\vec{u} &:= \vec{\phi} \times \vec{v} \\ &= \left[a^2u\sqrt{u^2 - 1}\sqrt{1 - v^2} \cos \phi, \quad a^2u\sqrt{u^2 - 1}\sqrt{1 - v^2} \sin \phi, \quad a^2(u^2 - 1)v \right]^T\end{aligned}$$

Taking the norm of $\vec{v}$, $\vec{\phi}$, and $\frac{\partial \vec{x}}{\partial u}$ give us the scale factors

$$\begin{aligned}h_u &= a\sqrt{\frac{u^2 - v^2}{u^2 - 1}} \\ h_v &= ||\vec{v}|| = a\sqrt{\frac{u^2 - v^2}{1 - v^2}} \\ h_\phi &= ||\vec{\phi}|| = a\sqrt{(u^2 - 1)(1 - v^2)}\end{aligned}$$

And the gradient operator is therefore

$$\nabla f = \frac{1}{h_u^2} f_u \vec{u} + \frac{1}{h_v^2} f_v \vec{v} + \frac{1}{h_\phi^2} f_\phi \vec{\phi}.$$

Additionally, the differential surface area of a spheroid with fixed u and a is

$$\begin{aligned}dS &= h_v h_\phi dv d\phi \\ &= a^2 \sqrt{(u^2 - 1)(u^2 - v^2)} dv d\phi.\end{aligned}$$

The Laplacian in spheroidal coordinates is

$$\Delta = \frac{1}{a^2(u^2 - v^2)} \left(\frac{\partial}{\partial u} \left[(u^2 - 1) \frac{\partial}{\partial u} \right] + \frac{\partial}{\partial v} \left[(1 - v^2) \frac{\partial}{\partial v} \right] \right) + \frac{1}{a^2(u^2 - 1)(1 - v^2)} \frac{\partial^2}{\partial \phi^2}$$

1.2 Useful Expressions in Spheroidal Coordinates

- The euclidean norm

$$||(u, v, \phi)|| = a^2(u^2 + v^2 - 1).$$

2 Spheroidal Harmonic Functions

The spheroidal surface harmonic functions are defined as

$$Y_n^m(v, \phi) = P_n^m(v) e^{im\phi} \quad (1)$$

where P_n^m is the associated Legendre function of the first kind with degree n and order m .

The spheroidal solid harmonic functions for the interior of the spheroid are

$$R_n^m(u, v, \phi) = P_n^m(u) Y_n^m(v, \phi) \quad (2)$$

And for the exterior of the spheroid,

$$I_n^m(u, v, \phi) = Q_n^m(u) Y_n^m(v, \phi) \quad (3)$$

where Q_n^m is the associated Legendre function of the **second** kind with degree n and order m .

Like spherical solid harmonics, R_n^m and I_n^m form the solution space of Laplace's equation in the interior and exterior of a spheroid (respectively).

3 Legendre functions off the cut

3.1 Overview

Associated Legendre functions of the first kind P_n^m and the second kind Q_n^m are solutions to the ODE

$$(1 - x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] y = 0.$$

In the application of prolate spheroidal harmonics, we need to evaluate $P_n^m(v)$ and $Q_n^m(v)$ for $-1 \leq v \leq 1$ (as in the surface harmonics) as well as for $P_n^m(u)$ and $Q_n^m(u)$ for $u > 1$. Algorithms to evaluate associated Legendre functions on $[-1, 1]$ are not stable for $(1, \infty)$, so we implement a different algorithm which will now be described:

Both P_n^m and Q_n^m obey the recurrence relation over n

$$(n - m + 1)P_{n+1}^m(x) - (2n + 1)xP_n^m(x) + (n + m)P_{n-1}^m(x) = 0, \quad n \geq 1 \quad (4)$$

We use the initial functions

$$\begin{aligned} P_m^m(x) &= (2m - 1)!!(x^2 - 1)^{m/2} \\ P_{m+1}^m(x) &= x(2m + 1)P_m^m(x) \end{aligned}$$

since they can be calculated directly. Using these with the forward recurrence 4 is a stable method to compute $P_{m+2}^m, \dots, P_N^m$ where the desired maximum degree is N . However, to compute the set of Q_n^m stably, we must use backward recurrence.

The Wronskian relationship is

$$W(P_n^m, Q_n^m) = \frac{(n+m)!(-1)^m}{(n-m)!(1-x^2)} \quad (5)$$

Combining the recurrence relation 4 and Wronskian 5 gives us a recurrence relation with both P_n^m and Q_n^m

$$P_n^m Q_{n-1}^m - P_{n-1}^m Q_n^m = \frac{(n+m-1)!}{(n-m)!} (-1)^m, \quad n \geq 1 \quad (6)$$

We need an additional equation that relates Q_{n-1}^m and Q_n^m . Dividing equation 4 by Q_n^m and rearranging gives us the continued fraction formula

$$H(x) = \frac{Q_n^m}{Q_{n-1}^m} = \frac{-\frac{n+m}{n-m+1}}{-\frac{2n+1}{n-m+1}x + \frac{Q_{n+1}^m}{Q_n^m}}, \quad n \geq 1 \quad (7)$$

Once P_{N-1}^m and P_N^m have been computed by a forward recurrence, we combine equations 6 and 7 to obtain

$$Q_{N-1}^m(x) = \frac{(n+m-1)!}{(n-m)!} \frac{(-1)^m}{P_N^m(x) - H(x)P_{N-1}^m(x)}$$

$$Q_N^m(x) = H(x)Q_{N-1}^m(x)$$

Now that we have the last two second kind legendre functions Q_N^m and Q_{N-1}^m , we can then use the original recurrence relation 4 to iterate backwards and obtain $Q_{N-2}^m, \dots, Q_m^m$.

In order to compute P_n^m and Q_n^m for $m < 0$, we use the formula

$$P_n^m(x) = \frac{(n+m)!}{(n-m)!} P_n^{|m|}(x), \quad m < 0$$

3.2 Evaluation of $H(x)$

To evaluate the continued fraction $H(x)$, we implemented the modified Lentz's algorithm as in [1]. The algorithm to evaluate

$$f = b_0 + \frac{a_1}{b_1 + \frac{a_2}{b_2 + \dots}}$$

is as follows:

Applying this to our continued fraction of interest

$$H(x) = \frac{Q_n^m(x)}{Q_{n-1}^m(x)} = \frac{-\frac{n+m}{n-m+1}}{-\frac{2n+1}{n-m+1}x + \frac{Q_{n+1}^m}{Q_n^m}} = \frac{-\frac{n+m}{n-m+1}}{-\frac{2n+1}{n-m+1}x + \frac{-\frac{(n+1)+m}{(n+1)-m+1}}{-\frac{2(n+1)+1}{(n+1)-m+1}x + \dots}}$$

Algorithm 1 Modified Lentz's Algorithm

```
 $f_0 \leftarrow b_0$ 
if  $b_0 = 0$  then
   $f_0 \leftarrow 10^{-30}$ 
end if
 $C_0 \leftarrow f_0$ 
 $D_0 \leftarrow 0$ 
for  $j = 1, 2, \dots$  do
   $D_j \leftarrow b_j + a_j D_{j-1}$ 
  if  $D_j = 0$  then
     $D_j \leftarrow 10^{-30}$ 
  end if
   $C_j \leftarrow b_j + a_j / C_{j-1}$ 
  if  $C_j = 0$  then
     $C_j \leftarrow 10^{-30}$ 
  end if
   $D_j \leftarrow 1 / D_j$ 
   $\Delta_j \leftarrow C_j D_j$ 
   $f_j \leftarrow \Delta_j f_{j-1}$ 
  if  $|\Delta_j - 1| < \epsilon$  then exit
  end if
end for
```

we have

$$\begin{aligned} b_0 &= 0 \\ a_j &= -\frac{(n+j-1)+m}{n+j-m}, \quad j \geq 1 \\ b_j &= -\frac{2(n+j-1)+1}{n+j-m}x, \quad j \geq 1 \end{aligned}$$

Modified Lentz's Algorithm 1 is particularly useful because it accounts for the case when $b_0 = 0$.

3.3 Derivative Computation

We need to evaluate the derivatives of $P_n^m(x)$ and $Q_n^m(x)$ for $x > 1$. There are several formulas to compute these, including

$$(1-x^2) \frac{dP_n^m}{dx} = (m-n-1)P_{n+1}^m + (n+1)xP_n^m \quad (8)$$

Q_n^m satisfies equation 8 as well.

4 Boundary Integral Operators

4.1 Laplace Double Layer

4.1.1 Formulas

For a spheroid with eccentricity $1/u_0$, define $a_n^m = \frac{(n-m)!}{(n+m)!}(-1)^m(u_0^2 - 1)$. The spheroidal surface harmonics Y_n^m diagonalize the Double layer operator:

$$D[Y_n^m](\vec{x}) = \lambda_n^m Y_n^m(\vec{x})$$

where

$$\lambda_n^m = \begin{cases} \frac{a_n^m}{2} [P_n^{m'}(u_0)Q_n^m(u_0) + P_n^m(u_0)Q_n^{m'}(u_0)], & x \in \Gamma \\ a_n^m P_n^{m'}(u_0)Q_n^m(u), & x \in \Omega^C \\ a_n^m Q_n^{m'}(u_0)P_n^m(u), & x \in \Omega \end{cases} \quad (9)$$

4.1.2 Double Layer Algorithm

To evaluate the double layer of a continuous density σ on the spheroid surface with eccentricity $1/u_0$, perform the following procedure:

1. Expand σ in the Y_n^m basis with a spheroidal harmonic transform $\sigma = \sum_{n,m} \sigma_n^m Y_n^m$.
2. Use the formulas (9) to compute $D[\sigma] = D \left[\sum_{n,m} \sigma_n^m Y_n^m \right] = \sum_{n,m} \sigma_n^m \lambda_n^m Y_n^m$
3. Perform an inverse spheroidal transform to recover $D[\sigma]$ in spatial spheroidal coordinates.

4.1.3 Derivation

Let $\sigma(v, \phi)$ be a continuous function on the spheroid surface Γ with fixed $u = u_0$. The order- p expansion of σ in the spheroidal harmonic basis is

$$\sigma = \sum_{n,m} \sigma_n^m Y_n^m(v, \phi).$$

Similarly, we expand $D[\sigma]$ in the exterior and interior to order p , using the exterior and interior spheroidal solid harmonics

$$\begin{aligned} D[\sigma]^i(u, v, \phi) &= \sum_{m,n} a_{nm}^i R_n^m(u, v, \phi) = \sum_{m,n} a_{nm}^i P_n^m(u) Y_n^m(v, \phi) & (\text{interior}) \\ D[\sigma]^e(u, v, \phi) &= \sum_{m,n} a_{nm}^e I_n^m(u, v, \phi) = \sum_{m,n} a_{nm}^e Q_n^m(u) Y_n^m(v, \phi) & (\text{exterior}) \end{aligned}$$

The jump conditions of the double layer are:

$$\begin{aligned} \lim_{u \rightarrow u_0^-} D[\sigma]^i(u, v, \phi) &= \frac{1}{2} \sigma(v, \phi) & (\text{interior}) \\ \lim_{u \rightarrow u_0^+} D[\sigma]^e(u, v, \phi) &= \frac{3}{2} \sigma(v, \phi) & (\text{exterior}) \end{aligned}$$

Subtracting these expressions gives us the jump condition between the interior and exterior

$$D[\sigma]^i(u_0, v, \phi) - D[\sigma]^e(u_0, v, \phi) = -\sigma(v, \phi).$$

Substituting the expansions above, we have

$$\sum_{m,n} a_{nm}^i P_n^m(u_0) Y_n^m(v, \phi) - \sum_{m,n} a_{nm}^e Q_n^m(u_0) Y_n^m(v, \phi) = -\sum_{n,m} \sigma_n^m Y_n^m(v, \phi)$$

Since Y_n^m are an orthogonal basis, we can match each term in the expansions, and we obtain

$$a_{nm}^i P_n^m(u_0) - a_{nm}^e Q_n^m(u_0) = -\sigma_n^m \quad (10)$$

We need an additional equation to solve for a_{nm}^i and a_{nm}^e . We use the fact that the normal derivative to the surface $D'[\sigma] := \frac{\partial}{\partial n} D[\sigma]$ is continuous at the spheroid surface:

$$\lim_{u \rightarrow u_0^-} D'[\sigma]^i(u, v, \phi) - \lim_{u \rightarrow u_0^+} D'[\sigma]^e(u, v, \phi) = 0$$

Recall that the normal derivative in spheroidal coordinates is

$$\frac{\partial}{\partial n} = \frac{1}{a} \sqrt{\frac{u^2 - 1}{u^2 - v^2}} \frac{\partial}{\partial u}.$$

Then, substituting the expansions we have

$$\begin{aligned} & \frac{1}{a} \sqrt{\frac{u_0^2 - 1}{u_0^2 - v^2}} \left[\lim_{u \rightarrow u_0^-} \frac{\partial}{\partial u} \sum_{m,n} a_{nm}^i P_n^m(u) Y_n^m(v, \phi) - \lim_{u \rightarrow u_0^+} \frac{\partial}{\partial u} \sum_{m,n} a_{nm}^e Q_n^m(u) Y_n^m(v, \phi) \right] = 0 \\ \Rightarrow & \frac{1}{a} \sqrt{\frac{u_0^2 - 1}{u_0^2 - v^2}} \left[\sum_{m,n} a_{nm}^i P_n^{m'}(u_0) Y_n^m(v, \phi) - \sum_{m,n} a_{nm}^e Q_n^{m'}(u_0) Y_n^m(v, \phi) \right] = 0 \\ \Rightarrow & \sum_{m,n} a_{nm}^i P_n^{m'}(u_0) Y_n^m(v, \phi) - \sum_{m,n} a_{nm}^e Q_n^{m'}(u_0) Y_n^m(v, \phi) = 0 \end{aligned}$$

Once again, by the orthogonality of the Y_n^m functions, we can match each coefficient and obtain

$$a_{nm}^i P_n^{m'}(u_0) - a_{nm}^e Q_n^{m'}(u_0) = 0 \quad (11)$$

Equations (10) and (11) give us a system of equations which we can use to solve for a_{nm}^i and a_{nm}^e . We have

$$\begin{bmatrix} P_n^m(u_0) & -Q_n^m(u_0) \\ P_n^{m'}(u_0) & -Q_n^{m'}(u_0) \end{bmatrix} \begin{bmatrix} a_{nm}^i \\ a_{nm}^e \end{bmatrix} = \begin{bmatrix} -\sigma_n^m \\ 0 \end{bmatrix}$$

Note that the determinant of the above matrix is simply the negative of the Wronskian, $-W(P_n^m, Q_n^m)(u_0) = \frac{(n+m)!(-1)^m}{(n-m)!(u_0^2-1)}$. Thus the solution is

$$\begin{aligned} \begin{bmatrix} a_{nm}^i \\ a_{nm}^e \end{bmatrix} &= -\frac{1}{W(P_n^m, Q_n^m)(u_0)} \begin{bmatrix} -Q_n^{m'}(u_0) & Q_n^m(u_0) \\ -P_n^{m'}(u_0) & P_n^m(u_0) \end{bmatrix} \begin{bmatrix} -\sigma_n^m \\ 0 \end{bmatrix} \\ &= -\frac{1}{W(P_n^m, Q_n^m)(u_0)} \begin{bmatrix} Q_n^{m'}(u_0) \sigma_n^m \\ P_n^{m'}(u_0) \sigma_n^m \end{bmatrix} \\ &= \frac{(n-m)!}{(n+m)!} (-1)^m (u_0^2 - 1) \sigma_n^m \begin{bmatrix} Q_n^{m'}(u_0) \\ P_n^{m'}(u_0) \end{bmatrix} \end{aligned}$$

Finally, plugging this into the original interior and exterior expansions gives us the desired result:

$$\begin{aligned} D[\sigma]^i(u, v, \phi) &= \sum_{m,n} \frac{(n-m)!}{(n+m)!} (-1)^m (u_0^2 - 1) \sigma_n^m Q_n^{m'}(u_0) P_n^m(u) Y_n^m(v, \phi) \quad (\text{interior}) \\ D[\sigma]^e(u, v, \phi) &= \sum_{m,n} \frac{(n-m)!}{(n+m)!} (-1)^m (u_0^2 - 1) \sigma_n^m P_n^{m'}(u_0) Q_n^m(u) Y_n^m(v, \phi) \quad (\text{exterior}) \end{aligned}$$

Equivalently: For any Y_n^m ,

$$\begin{aligned} D[Y_n^m]^i &= \frac{(n-m)!}{(n+m)!} (-1)^m (u_0^2 - 1) Q_n^{m'}(u_0) P_n^m(u) Y_n^m(v, \phi) \quad (\text{interior}) \\ D[Y_n^m]^e &= \frac{(n-m)!}{(n+m)!} (-1)^m (u_0^2 - 1) P_n^{m'}(u_0) Q_n^m(u) Y_n^m(v, \phi) \quad (\text{exterior}) \end{aligned}$$

4.2 Laplace Single Layer

4.2.1 Eigenvalue Formulas

Unlike the double layer operator, the single layer operator S is not diagonalized by Y_n^m . It is, however, diagonalized by $\frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$. We have

$$S \left[\frac{Y_n^m}{\sqrt{u_0^2 - v^2}} \right] (\vec{x}) = \lambda_n^m Y_n^m(\vec{x})$$

The eigenvalues are

$$\lambda_n^m = \begin{cases} c_n^m P_n^m(u) Q_n^m(u_0), & x \in \Omega^C \\ c_n^m P_n^m(u_0) Q_n^m(u), & x \in \Omega \end{cases} \quad (12)$$

where $c_n^m = a \frac{(n-m)!}{(n+m)!} (-1)^m \sqrt{u_0 - 1}$. S is continuous across the boundary of the spheroid, so we can simply use either the interior or exterior formula and plug in $u = u_0$ to obtain

$$\lambda_n^m = c_n^m P_n^m(u_0) Q_n^m(u_0), \quad x \in \Gamma.$$

4.2.2 Change of Basis, $Y_n^m \rightarrow \frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$

We will see in the next section that evaluating the single layer of some arbitrary surface density σ requires a change of basis from Y_n^m to $\frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$. Let $\tilde{\sigma}_n^m$ be the coefficients of $\frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$, that is

$$\sigma = \sum_{n,m} \tilde{\sigma}_n^m \frac{Y_n^m}{\sqrt{u_0^2 - v^2}}.$$

Then coefficients in the Y_n^m basis are

$$\sigma_n^m = \sum_{n',m'} \tilde{\sigma}_{n'}^{m'} \left\langle Y_n^m, \frac{Y_{n'}^{m'}}{\sqrt{u_0^2 - v^2}} \right\rangle$$

Equivalently, we can define the change-of-basis matrix $G : \frac{Y_n^m}{\sqrt{u_0^2 - v^2}} \rightarrow Y_n^m$ as

$$G_{ij} = \left\langle Y_{n_i}^{m_i}, \frac{Y_{n_j}^{m_j}}{\sqrt{u_0^2 - v^2}} \right\rangle \quad (13)$$

Thus, to transform from Y_n^m to $\frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$, we compute the inverse of G .

4.2.3 Single Layer Algorithm

To actually evaluate the single layer of a continuous density on the spheroid surface, σ , we perform the following procedure:

1. Expand σ in the Y_n^m basis with a spheroidal harmonic transform: $\sigma = \sum_{n,m} \sigma_n^m Y_n^m$
2. Perform a change of basis $Y_n^m \rightarrow \frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$ by computing $G^{-1} \sigma_n^m$. We then have coefficients $\tilde{\sigma}_n^m$ in the new basis: $\sigma = \sum_{n,m} \tilde{\sigma}_n^m \frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$.
3. Use the above formulas (12) to compute $S \left[\sum_{n,m} \tilde{\sigma}_n^m \frac{Y_n^m}{\sqrt{u_0^2 - v^2}} \right] = \sum_{n,m} \tilde{\sigma}_n^m \lambda_n^m Y_n^m$.
4. Now we have the coefficients of $S[\sigma]$ in the Y_n^m basis, which are $\tilde{\sigma}_n^m \lambda_n^m$. Perform an inverse spheroidal harmonic transform to recover $S[\sigma]$.

4.2.4 Derivation

The derivation procedure largely mirrors that of the double layer. One major difference is that S is diagonalized by $\frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$. Thus, given a function σ on the spheroid surface, we consider its expansion in the $\frac{Y_n^m}{\sqrt{u_0^2 - v^2}}$ basis. We again write $S[\sigma]$ in the interior and exterior in the spheroidal solid harmonic basis. We therefore

use the following expansions:

$$\begin{aligned}\sigma(v, \phi) &= \sum_{n,m} \tilde{\sigma}_n^m \frac{Y_n^m(v, \phi)}{\sqrt{u^2 - v^2}} \\ S[\sigma]^i(u, v, \phi) &= \sum_{n,m} c_{nm}^i P_n^m(u) Y_n^m(v, \phi) \\ S[\sigma]^e(u, v, \phi) &= \sum_{n,m} c_{nm}^e Q_n^m(u) Y_n^m(v, \phi)\end{aligned}$$

S is continuous approaching the surface from the interior and exterior, so we have

$$S[\sigma]^i(u_0, v, \phi) - S[\sigma]^e(u_0, v, \phi) = 0.$$

Substituting the expansions and matching the coefficients on each Y_n^m basis function gives us

$$c_{nm}^i P_n^m(u_0) - c_{nm}^e Q_n^m(u_0) = 0 \quad (14)$$

The normal derivative of S , S' , has a jump condition similar to that of D :

$$\begin{aligned}\lim_{u \rightarrow u_0^-} S'[\sigma]^i(u, v, \phi) &= \frac{3}{2} \sigma(v, \phi) \quad (\text{interior}) \\ \lim_{u \rightarrow u_0^+} S'[\sigma]^e(u, v, \phi) &= \frac{1}{2} \sigma(v, \phi) \quad (\text{exterior})\end{aligned}$$

Subtracting these expressions, we have

$$\lim_{u \rightarrow u_0^-} S'[\sigma]^i(u, v, \phi) - \lim_{u \rightarrow u_0^+} S'[\sigma]^e(u, v, \phi) = \sigma(v, \phi).$$

Substituting the spheroidal harmonic expansions into this equation gives us

$$\begin{aligned}\frac{1}{a} \sqrt{\frac{u_0^2 - 1}{u_0^2 - v^2}} \left[\sum_{n,m} c_{nm}^i P_n^{m'}(u_0) Y_n^m(v, \phi) - \sum_{n,m} c_{nm}^e Q_n^{m'}(u_0) Y_n^m(v, \phi) \right] &= \sum_{n,m} \tilde{\sigma}_n^m \frac{Y_n^m(v, \phi)}{\sqrt{u_0^2 - v^2}} \\ \Rightarrow \frac{\sqrt{u_0^2 - 1}}{a} \left[\sum_{n,m} c_{nm}^i P_n^{m'}(u_0) Y_n^m(v, \phi) - \sum_{n,m} c_{nm}^e Q_n^{m'}(u_0) Y_n^m(v, \phi) \right] &= \sum_{n,m} \tilde{\sigma}_n^m Y_n^m(v, \phi)\end{aligned}$$

Now, since $\{Y_n^m\}$ is an orthogonal basis, we can match the coefficients for each Y_n^m , and we get

$$c_{nm}^i P_n^{m'}(u_0) - c_{nm}^e Q_n^{m'}(u_0) = \frac{a}{\sqrt{u_0^2 - 1}} \tilde{\sigma}_n^m \quad (15)$$

Like the double layer derivation, we now have two equations (14) and (15) which we can use to solve for c_{nm}^i and c_{nm}^e . The linear system is

$$\begin{bmatrix} P_n^m(u_0) & -Q_n^m(u_0) \\ P_n^{m'}(u_0) & -Q_n^{m'}(u_0) \end{bmatrix} \begin{bmatrix} c_{nm}^i \\ c_{nm}^e \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{a}{\sqrt{u_0^2 - 1}} \tilde{\sigma}_n^m \end{bmatrix}$$

This is the same matrix as in the double layer, case, so we reuse the inverse calculated in that section:

$$\begin{aligned}\begin{bmatrix} c_{nm}^i \\ c_{nm}^e \end{bmatrix} &= -\frac{1}{W(P_n^m, Q_n^m)(u_0)} \begin{bmatrix} -Q_n^{m'}(u_0) & Q_n^m(u_0) \\ -P_n^{m'}(u_0) & P_n^m(u_0) \end{bmatrix} \begin{bmatrix} 0 \\ \frac{a \tilde{\sigma}_n^m}{\sqrt{u_0^2 - 1}} \end{bmatrix} \\ &= -\frac{1}{W(P_n^m, Q_n^m)(u_0)} \cdot \frac{a \tilde{\sigma}_n^m}{\sqrt{u_0^2 - 1}} \begin{bmatrix} Q_n^m(u_0) \\ P_n^m(u_0) \end{bmatrix} \\ &= \frac{(n-m)!}{(n+m)!} (-1)^m a \tilde{\sigma}_n^m \sqrt{u_0^2 - 1} \begin{bmatrix} Q_n^m(u_0) \\ P_n^m(u_0) \end{bmatrix}\end{aligned}$$

Thus, we obtain the desired formulas:

$$S[\sigma]^i(u, v, \phi) = \sum_{n,m} \frac{(n-m)!}{(n+m)!} (-1)^m a \tilde{\sigma}_n^m \sqrt{u_0^2 - 1} Q_n^m(u_0) P_n^m(u) Y_n^m(v, \phi) \quad (\text{interior})$$

$$S[\sigma]^e(u, v, \phi) = \sum_{n,m} \frac{(n-m)!}{(n+m)!} (-1)^m a \tilde{\sigma}_n^m \sqrt{u_0^2 - 1} P_n^m(u_0) Q_n^m(u) Y_n^m(v, \phi) \quad (\text{exterior})$$

Note that $\tilde{\sigma}_n^m$ are the coefficients of σ in the $Y_n^m / \sqrt{u_0^2 - v^2}$ basis.

For a specific Y_n^m , we have

$$S \left[\frac{Y_n^m}{\sqrt{u_0^2 - v^2}} \right]^i (u, v, \phi) = \frac{(n-m)!}{(n+m)!} (-1)^m a \sqrt{u_0^2 - 1} Q_n^m(u_0) P_n^m(u) Y_n^m(v, \phi) \quad (\text{interior})$$

$$S \left[\frac{Y_n^m}{\sqrt{u_0^2 - v^2}} \right]^e (u, v, \phi) = \frac{(n-m)!}{(n+m)!} (-1)^m a \sqrt{u_0^2 - 1} P_n^m(u_0) Q_n^m(u) Y_n^m(v, \phi) \quad (\text{exterior})$$

5 Multiple Particles

5.1 Position and Rotation Tracking

Consider a system of M particles $\{\Gamma_i\}_{i=1}^M$. Each particle has a center $\mathbf{c}_i$ and has been rotated from its original upright position by an angle θ_i around the y -axis and then an angle ϕ_i around the z -axis. That is, given a point $\mathbf{x} \in \Gamma_0$ where Γ_0 is a spheroid with the same shape as Γ_i but centered at the origin and unrotated, the corresponding point on Γ_i is

$$\begin{aligned} \mathbf{x}' &= R_z(\phi_i) R_y(\theta_i) \mathbf{x} + \mathbf{c}_i \\ &= \begin{bmatrix} \cos(\theta_i) \cos(\phi_i) & -\sin(\phi_i) & \sin(\theta_i) \cos(\phi_i) \\ \cos(\theta_i) \sin(\phi_i) & \cos(\phi_i) & \sin(\theta_i) \sin(\phi_i) \\ -\sin(\theta_i) & 0 & \cos(\theta_i) \end{bmatrix} \mathbf{x} + \mathbf{c}_i \end{aligned}$$

Thus, to restore Γ_i to the default centered position, we apply

$$\mathbf{x} = R_y^T(\theta_i) R_z^T(\phi_i) (\mathbf{x}' - \mathbf{c}_i)$$

It will be convenient and useful to denote the "orientation vector" $\mathbf{v}$, which points parallel to the spheroid's axis of rotational symmetry, in the direction corresponding to $v = -1$. For example, on a default spheroid centered at the origin, $\mathbf{v} = [0, 0, 1]^T$. Since spheroids are rotationally symmetric, this vector completely describes the orientation of a spheroid. For angles of rotation θ_i and ϕ_i , the orientation will be

$$\begin{aligned} \mathbf{v}_i &= R_z(\phi_i) R_y(\theta_i) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \sin \theta_i \cos \phi_i \\ \sin \theta_i \sin \phi_i \\ \cos \theta_i \end{bmatrix} \end{aligned}$$

5.2 Approximating Distance Between Spheroids

For multiparticle simulations, we want to know the distance between all pairs of particles to group them into "neighbors", which are close enough that using spheroidal harmonics to compute layer potentials is desired, or far enough away that smooth quadrature (which is much less expensive) is sufficiently accurate. Unfortunately, there is no closed-form expression for the distance between two spheroids. I propose the following simple and inexpensive method to approximate the distance:

Given two spheroids Γ_1 and Γ_2 with parameters a_1, u_1, a_2, u_2 have some fixed displacement $\mathbf{c}_2 - \mathbf{c}_1$ between their centers, the closest distance between each surface is determined by their relative orientations. If $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular to the displacement, then the spheroid surfaces are "as far away from each other as possible" given that the displacement stays constant. The displacement is in line with the spheroids' semi-minor axes, so the distance is

$$\|\mathbf{c}_2 - \mathbf{c}_1\| - a_1\sqrt{u_1^2 - 1} - a_2\sqrt{u_2^2 - 1}$$

i.e., equivalent to the distance between **spheres** centered at the same point as Γ_1 and Γ_2 with radii equal to the corresponding **semi-minor** axes of each spheroid.

On the other hand, if $\mathbf{v}_1$ and $\mathbf{v}_2$ are both parallel to $\mathbf{c}_2 - \mathbf{c}_1$, they are as close as they could possibly be to each other because the displacements lines up with their semi-major axes. In this case the distance is

$$\|\mathbf{c}_2 - \mathbf{c}_1\| - a_1u_1 - a_2u_2$$

i.e., equivalent to the distance between spheres centered at the same point as Γ_1 and Γ_2 with radii equal to the corresponding **semi-major** axes of each spheroid.

We obtain similar-looking results by considering when spheroid 1's orientation is parallel to displacement and spheroid 2's orientation is perpendicular. Then we use spheroid 1's semi-major axis and spheroid 2's semi-minor axis (and vice-versa). Let $\tilde{d}$ be the approximated distance between these spheroid surfaces. Then we should expect

$$\tilde{d} = \begin{cases} \|\mathbf{c}_2 - \mathbf{c}_1\| - a_1\sqrt{u_1^2 - 1} - a_2\sqrt{u_2^2 - 1} & \text{if } \mathbf{v}_1 \cdot (\mathbf{c}_2 - \mathbf{c}_1) = \mathbf{v}_2 \cdot (\mathbf{c}_2 - \mathbf{c}_1) = 0 \\ \|\mathbf{c}_2 - \mathbf{c}_1\| - a_1u_1 - a_2u_2 & \text{if } \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} \cdot \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|} = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} \cdot \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|} = 1 \\ \|\mathbf{c}_2 - \mathbf{c}_1\| - a_1\sqrt{u_1^2 - 1} - a_2u_2 & \text{if } \mathbf{v}_1 \cdot (\mathbf{c}_2 - \mathbf{c}_1) = 0, \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} \cdot \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|} = 1 \\ \|\mathbf{c}_2 - \mathbf{c}_1\| - a_1u_1 - a_2\sqrt{u_2^2 - 1} & \text{if } \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} \cdot \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|} = 1, \mathbf{v}_2 \cdot (\mathbf{c}_2 - \mathbf{c}_1) = 0 \end{cases}$$

To account for all other cases, use a simple linear interpolation of the "alignment", $t_i = \left| \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|} \cdot \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|} \right|$, which satisfies $0 \leq t_i \leq 1$.

$$\tilde{d} = \|\mathbf{c}_2 - \mathbf{c}_1\| - \left[t_1a_1u_1 + (1 - t_1)a_1\sqrt{u_1^2 - 1} \right] - \left[t_2a_2u_2 + (1 - t_2)a_2\sqrt{u_2^2 - 1} \right]$$

Geometrically, this means that we approximate the spheroids as spheres whose radii depend on how closely their orientations align with the displacement between each spheroid.¹

¹Interestingly, instead using $t_i = \left| \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|} \cdot \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|} \right|^2$ is a much better approximation of the true distance between spheroids. This heuristic turns out to have a relative accuracy of at worst about 10^{-2} . However, it is not strictly less than the true distance, and so it sometimes underestimates the closeness of two particles, which is highly undesirable.

References

- [1] W. Press. *Numerical Recipes in Fortran 77: The Art of Scientific Computing*. Cambridge: Cambridge University Press, 1997.